NOTE An Access Token has a validity period configured in Keycloak, and it should be as short as possible to reduce the exploitation time window in case the token gets leaked. An acceptable length is 5 minutes. When the token expires, the OAuth2 Client can ask the Authorization Server for a new one using a third type of token called a *Refresh Token* (which also has a validity period). The refresh mechanism is handled by Spring Security transparently, and I won't describe it further.

ADOPTING THE TOKEN RELAY PATTERN IN SPRING CLOUD GATEWAY

Spring Cloud Gateway implements the token relay pattern as a filter. In the Edge Service project (edge-service), open the application.yml file and add `TokenRelay` as a default filter, since we want it applied to all routes.

Listing 12.1 Enabling the token relay pattern in Spring Cloud Gateway

```
spring:
  cloud:
    gateway:
      default-filters:
        - SaveSession
        - TokenRelay      ❶
```

❶ Enables the propagation of the Access Token when calling a downstream service

With the filter enabled, Spring Cloud Gateway takes care of propagating the right Access Token as an `Authorization` header in all outgoing requests to Catalog Service and Order Service. For example:

```
GET /orders
Authorization: Bearer <access_token>
```

NOTE Unlike ID Tokens which are JWTs, the OAuth2 framework doesn't enforce a data format for Access Tokens. They can be of any String-based form. The most popular format is JWT, though, so that's how we'll parse Access Tokens on the consumer side (Catalog Service and Order Service).